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Pedicle Morphology of the Lower Thoracic and Lumbar Spine in a Chinese Population

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Knowledge of pedicle diameter and surface landmarks is crucial for safe placement of screws. Little attention has been paid to variations of entrance points for pedicle screws, differentiation of male and female pedicle sizes and pedicle size differences in nonwhite populations. Forty thoracolumbar spinal columns from T9 to L5 were measured using vernier calipers. Cephalad-caudad and medial-lateral diameter of the pedicle, length of the pedicle from posterior cortex to anterior cortex at the midline and parallel to midline was measured. Relation of the centre of the pedicle to the transverse process (TP) and to the superior facet joint was noted. Twenty-five male and 15 female specimens were measured. Average pedicle width in the female was 5.2 mm at T9 (SD 0.9) to 13 mm at L5 (SD 2.7) and in the male 6.0 mm at T9 (SD 1.1) to 12.8 mm at L5 (SD 2.7). Cephalad caudad diameter was 12.5 mm (SD 1.2) at T9 to 20.5 mm (SD 3.6) at L5 in the male and in the female 12.2 mm (SD 1.3) at T9 to 18.7 mm (SD3.9) at L5. All specimens had starting points cephalad to the midpoint of the TP at T9. At L5, 37 of 40 specimens had starting points at the midpoint of the TP. Starting points were parallel to the middle or lateral third of the superior facet joint at T9. At L5 starting points were at least one third of the facet joint lateral to the lateral border of the facet. Female pedicle width was smaller than male at T9 ($P = 0.03$) and T12 ($P = 0.04$). All pedicle widths except T11 are significantly smaller than the white population as measured by Zindrick et al. Male pedicle widths are significantly smaller than whites as measured by Olsewski et al. Only L5 is smaller than the white population as measured by Berry et al. Starting points should be more lateral in the low lumbar spine. Pedicle screw insertion depends on the sex and size of the patient, and the location of the vertebrae. [Key words: pedicle, morphology, gender differences, population differences]

Pedicle screw fixation of the spine is considered one of the most stable and versatile methods of fixation of the spine. With the initial methods of pedicular fixation described in the European literature, screw diameters were small and pedicle morphology was less critical to the application of these methods.[3,4,6,7] With the recent increased interest in internal fixation of the spine, knowledge of the size and position of the pedicle are crucial for safe application of these systems.[1,2,7–11] Description of the anatomy of the pedicles has been performed based on direct anatomic and radiologic measures of normal samples.[1,8,10] Upper and lower ranges of size have been reported. These have generally been reported on a white population, and have dictated many of the decisions in instrument and screw design.

The starting point of pedicle screws is of great importance to safe placement of internal fixation. Roy-Camille has advocated the junction of the mid-transverse process line and the mid-facet line as being the optimal position for placement of the pedicle screw.[6] Magerl describes the entry point as being at the intersection of the lateral border of the superior facet and the mid-transverse process.[4] Weinstein et al[9] advocated a more lateral approach to the pedicle and described entrance at the "nape of the neck" of the superior articular facet. The rationale for this approach was to prevent impingement of the adjoining articular process, although an added benefit was a perceived increased accuracy of pedicle instrumentation. Although there is acknowledged variability in the morphology of different levels of the spine, most authors advocate a universal method of screw placement and starting point.[2,4,6] This would seem to be contrary to the goal of placing screws accurately within the vertebral pedicles.

It is the purpose of this study to outline the anatomy of the lower thoracic and lumbar spine in an oriental population. We will also compare the differences encountered between the male and female population. Size and position of the pedicle are measured and optimal position of the center of the vertebral pedicle is defined. Although all pedicle screw systems vary in their characteristics, safe placement of screws is dependent on accurate anatomical knowledge.

Materials and Methods

Disarticulated vertebral bones from the Anatomy Department of the 4th Military Medical Institute in Xian City, China were used in this study. Forty sets of normal spinal columns from T9 to L5 were measured by one individual (Y.S).

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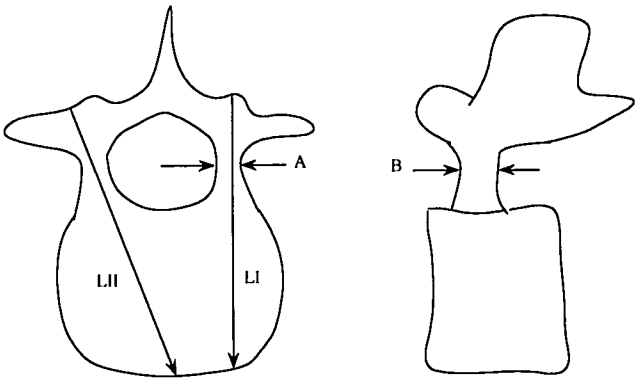


Figure 1. Anatomic diagram of points measured: Medial lateral width of pedicle (A), Cephalocaudal height of pedicle (B), Length I depth of parallel to sagittal plane of vertebral body, Length II, depth of body from posterior cortex to anterior midline cortex.

Measurement was performed using Vernier calipers with a resolution of 0.1 mm. Twenty-five of the specimens were male and 15 were female. The vertebrae were examined grossly and did not have bony disease or deformity. Measurements included the following:

Pedicle medial-lateral width (A) and cephalo-caudal diameter (B) taken at the narrowest section in the mid-point of the pedicle.

Length of pedicle as measured from the posterior cortex to the anterior vertebral cortex parallel to the sagittal diameter of the vertebral body (L1).

Length of pedicle measured from the posterior cortex to the mid-point of the anterior vertebral cortex (LII) (Figure 1).

The starting point (centre of the pedicle) was visually related to surface landmarks readily found on the posterior surface of the lamina. Visual examination in the lateral plane allowed estimation of the centre of the narrowest cephalad-caudad diameter of the pedicle. Notes were made as to whether this point lay at the top (T), upper half (U), midpoint (M), or lower half (B) of the transverse process (Figure 2). The medial-lateral mid-point of the pedicle was determined visually by examining the center of the narrowest diameter of the pedicle from cephalad along the axis of the neural canal.

The superior facet of the vertebrae was divided into three vertical segments by four vertical lines. A fourth segment was estimated by constructing a vertical line parallel to the other lines and equal in width to the segments of the facet joint. The center of the pedicle was related to these arbitrary regions of the facet (Figure 2). A further 10 specimens had measurements taken of the inclination of the pedicle in relation to the sagittal plane of the vertebral body.

Data were entered into a statistical program on a personal computer (Apple Macintosh, Cupertino California) and appropriate statistical analysis was performed using Statview SE (Abacus Concepts Software). Differences in means was determined using a two-tailed Student t test with a significance level of $P \leq 0.05$.

■ Results

Six measurements were obtained for each vertebrae from T9 to L5. A total of 40 specimens were measured. The

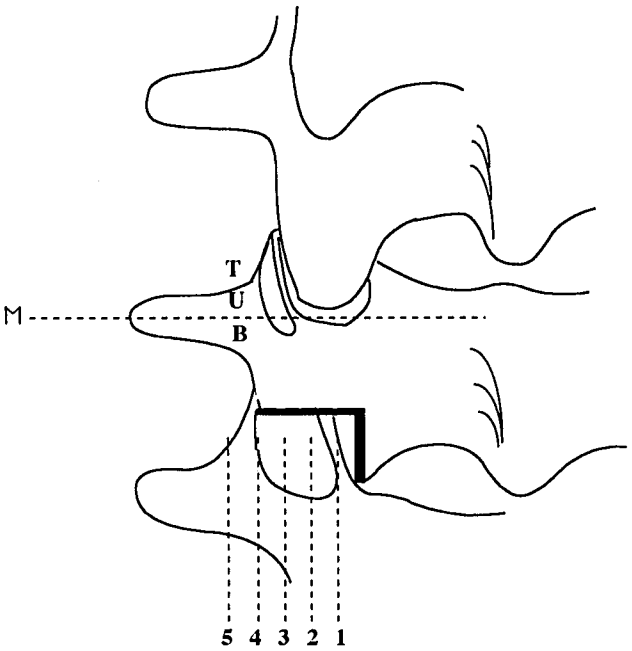


Figure 2. Posterior landmarks of starting point of pedicle: M = mid-transverse process, T = Top of transverse process, U = upper half of Transverse process, B = bottom half of transverse process. Facet joint divided into three segments by four vertical lines. Position of pedicle in medial-lateral plane related to zone of facet. Line 5 is one third of a facet diameter lateral to the facet joint.

total number of data points was 1,440. Cumulative summaries of the figures are listed in Tables 1, 2, and 3 with associated standard deviations.

Pedicle Width Medial-lateral and Cephalo-caudad

Medial lateral diameter (width) of the pedicle ranged from 5.2 mm (SD 0.9) at T9 to 13.0 mm (SD 2.7) at L5 in the female (Figure 3A). In the male, width ranged from 6.0 mm (SD 1.1) at T9 to 12.8 mm (SD 2.7) at L5 (Figure 3B). The L1 vertebral pedicle was consistently smaller than the pedicles above and below in each spinal column. Male to female differences were statistically significant at T9 ($P = 0.031$) and T12 ($P = 0.043$) (Table 1). There was no statistical difference between sexes in the lumbar spine. Pedicle cephalo-caudad mean diameters varied from 12.2 mm (SD 1.3) at T9 to 18.7 (SD 3.9) at L5 in the female (Figure 4A). The pedicle of L3 was narrowest in the lumbar spine. Male pedicle height ranged from 12.5 mm (SD 1.2) at T9 to 20.5 mm (SD 3.6) at L5 (Figure 4B). The narrowest diameter occurred at L3, with a diameter of 15 mm (SD 1.7). There was no statistically significant difference between males and females except at T12, where male pedicles were larger than females ($P = 0.005$) (Table 2).

Pedicle Depth

Pedicle depth (Length 1) measured parallel to midline from posterior cortex to anterior cortex are as noted. There was a significant difference between male and female at L1 (50.7 mm SD3.4 and 47.6 SD2.3) (P =

Table 1. Measurement of Pedicle Width (A) (mm)

Level	Current Study			Olsewski et al		Berry et al	Zindrick et al
	Male	Female	Cumulative	Male	Female		
T9	6.0 ± 1.1*	5.2 ± 0.9*	5.7 ± 1.1				6.1 ± 1.5‖
T10	7.0 ± 1.3	6.4 ± 1.4	6.8 ± 1.4				6.3 ± 1.7‖
T11	8.6 ± 1.7	7.5 ± 1.7	8.1 ± 1.8				7.8 ± 2.0
T12	8.8 ± 1.6*	7.5 ± 2.3*	8.3 ± 1.9			7.7 ± 2.1	7.1 ± 2.3‖
L1	7.2 ± 1.3†	6.6 ± 1.0‡	7.0 ± 1.2	9.5 ± 2.9†	7.7 ± 1.9‡	7.0 ± 1.9	8.7 ± 2.3‖
L2	7.6 ± 1.2†	7.1 ± 1.5	7.4 ± 1.3	9.6 ± 2.2†	7.9 ± 1.9	7.4 ± 1.6	8.9 ± 2.2‖
L3	9.4 ± 1.6†	9.0 ± 1.8	9.2 ± 1.6	11.7 ± 2.5†	9.6 ± 2.4	9.2 ± 1.3	10.3 ± 2.6‖
L4	10.8 ± 1.4†	10.2 ± 2.2‡	10.5 ± 1.8	14.7 ± 2.7†	12.5 ± 2.3‡	10.3 ± 1.6	12.9 ± 2.1‖
L5	12.8 ± 2.7†§	13.0 ± 2.7‡§	12.9 ± 2.7	21.1 ± 3.4†	18.4 ± 3.6‡	10.9 ± 3.4§	18.0 ± 4.1‖

0.003). (Figure 5A and B) The general trend is for pedicle length to decrease from L1 to L5.

Pedicle depth (Length II) measured from posterior cortex to midline along the pedicle is illustrated in the table. Males and females had significant differences at L1 ($P = 0.005$), with females smaller than males (Table 3).

Starting Point

The position of the starting point in relation to the transverse process is as described in Figure 6. At the lower thoracic region, the starting point is above the mid-point of the transverse process and as one goes distally, the starting point becomes more consistently at the mid-point of the TP.

Medial-lateral position begins at the mid-portion of the facet joint at T9 and progressively moves laterally. At the L5 level, almost all starting points are well lateral to the lateral border of the facet joints. Average distance lateral to the most lateral portion of the facet joint at L5 is 4.9 mm (SD 2). Inclination of pedicle is as noted in Figure 7.

■ Discussion

Spinal pedicle fixation has become a popular method of internal fixation of the spine. Many authors have described the morphometric aspects of the lower thoracic and lumbar spine; however, little emphasis has been placed on the specific differences between males and females in relation to sizes of pedicle. This would tend to confirm Olsewski's findings.[5] We have seen that in our sample, there were some significant differences in the size of the pedicle at the T12 level. This would have importance when considering stabilization of these segments. Given a choice, one should use a smaller-diameter pedicle screw at these upper lumbar and lower thoracic levels. It is interesting to note that Weinstein et al had a 40% failure rate when attempting screw insertion in the upper lumbar spine with a 7.0-mm screw. Changing screws to 5.5-mm diameter significantly improved success at these levels. The L1 pedicle is the smallest vertebral pedicle and requires added care when instrumenting this level. Minimal sizes in males were in the low 5-mm range, and it would seem prudent to place a 5-mm screw in this region. Minimal pedicle widths were smaller in the lower thoracic spine and in females with sizes of 3 mm and 4.2 mm at T9 and T10. This would have obvious consequences for insertion of pedicle screws. Pedicle heights were always greater than pedicle width and were never smaller than 9.2 mm in the male. In the female this measurement was never less than 9.3 mm. Unfortunately, height and weight were not available for these individuals.

Depth of pedicle parallel to the midline was greatest in the upper lumbar spine, with averages of 50.3 mm and 48.6 mm at L1 in males and females, respectively. Pedicle length shortens as one proceeds distally to the

Table 2. Pedicle Height (B) (mm)

Level	Current Study			Olsewski et al		Berry et al	Zindrick et al
	Male	Female	Cumulative	Male	Female		
T9	12.5 ± 1.2	12.2 ± 1.3	12.9 ± 1.2‖				13.8 ± 1.3‖
T10	14.4 ± 1.3	14.2 ± 1.7	14.3 ± 1.4‖				15.2 ± 2.0‖
T11	16.4 ± 1.6	16.0 ± 1.6	16.2 ± 1.6‖				17.4 ± 2.5‖
T12	17.1 ± 1.3*	15.9 ± 1.1*	16.6 ± 1.4§‖			17.2 ± 1.6§	15.8 ± 2.4‖
L1	15.9 ± 1.4†	15.5 ± 1.5	15.8 ± 1.4	17.0 ± 1.7†	15.3 ± 2.0	15.6 ± 1.4	15.4 ± 2.8
L2	15.4 ± 1.6	14.7 ± 1.0‡	15.1 ± 1.5	16.0 ± 2.2	15.3 ± 2.0‡	15.4 ± 1.0	15.0 ± 1.5
L3	15.3 ± 1.7†	14.3 ± 1.4	14.7 ± 1.6	16.1 ± 2.1†	15.0 ± 1.5	14.6 ± 1.2	14.9 ± 2.4
L4	15.3 ± 1.9*†	14.5 ± 2.6*	15.0 ± 2.2§	16.4 ± 2.5†	14.9 ± 1.5	13.0 ± 1.3§	14.8 ± 2.1
L5	20.5 ± 3.6†	18.7 ± 3.9‡	19.8 ± 3.7‖	17.4 ± 3.4†	16.2 ± 1.9‡	13.8 ± 2.5	14.0 ± 2.3‖

Table 3. Pedicle Length LI and LII (mm)

Level	LI			LII		
	Male	Female	Cumulative	Male	Female	Zindrick et al
T9	39.2 ± 3.2	37.7 ± 2.7	38.6 ± 3.0‖	43.6 ± 2.9	41.9 ± 3.0	41.2 ± 5.3‖
T10	39.4 ± 3.0	37.6 ± 2.5	38.8 ± 2.9‖	44.2 ± 2.5	47.7 ± 2.7	40.1 ± 4.6‖
T11	39.6 ± 3.1	38.0 ± 4.3	39.0 ± 3.6‖	44.6 ± 3.2	43.6 ± 4.6	41.2 ± 3.8‖
T12	43.5 ± 4.9	41.9 ± 5.8	42.9 ± 5.2	49.1 ± 5.3	47.8 ± 5.7	43.2 ± 5.0
L1	50.7 ± 3.4*	47.6 ± 2.3*	49.5 ± 3.4‖	55.6 ± 3.2*	52.7 ± 2.7*	44.7 ± 4.5‖
L2	50.3 ± 3.9	48.6 ± 2.2	49.7 ± 3.4‖	55.4 ± 3.8	53.2 ± 2.3	45.5 ± 3.7‖
L3	49.7 ± 3.8	48.9 ± 2.6	49.4 ± 3.4‖	55.6 ± 3.5	53.7 ± 2.6	44.4 ± 5.1‖
L4	47.9 ± 3.9	46.3 ± 2.4	47.3 ± 3.4‖	53.9 ± 4	51.8 ± 2.3	40.7 ± 4.3‖
L5	43.9 ± 4.1	43.1 ± 4.9	43.6 ± 4.4‖	51.9 ± 3.9	50.6 ± 4.5	33.7 ± 5.6‖

L5 vertebrae. This is similar to data published by Berry et al.[1] Depending on inclination of the pedicle screw, length should be limited to 45 mm in the lumbar spine.

Position of starting point varied to a great extent as one proceeds from cephalad to caudal. The general trend is for the starting point to begin more medial and proximal to the transverse process midline and then to gradually approach the mid-transverse pedicle level, as well as move well lateral to the articular surface of the superior facet. Weinstein et al[9] noted this trend and thought that the Roy-Camille method was superior in the lower thoracic and upper lumbar spine, whereas their more lateral method was more accurate in the lower lumbar spine. Migration of starting point requires varying starting point as successively distal levels of the spine are instrumented. One universal starting point would not be appropriate for all levels of lower thoracic and lumbar spine.

Angle of inclination of the pedicle to the midline is similar to previously reported values (Figure 7).[1,2,10,11] Progression distally produces more inclination away from the midline and preparation of the pedicle must take this into account.

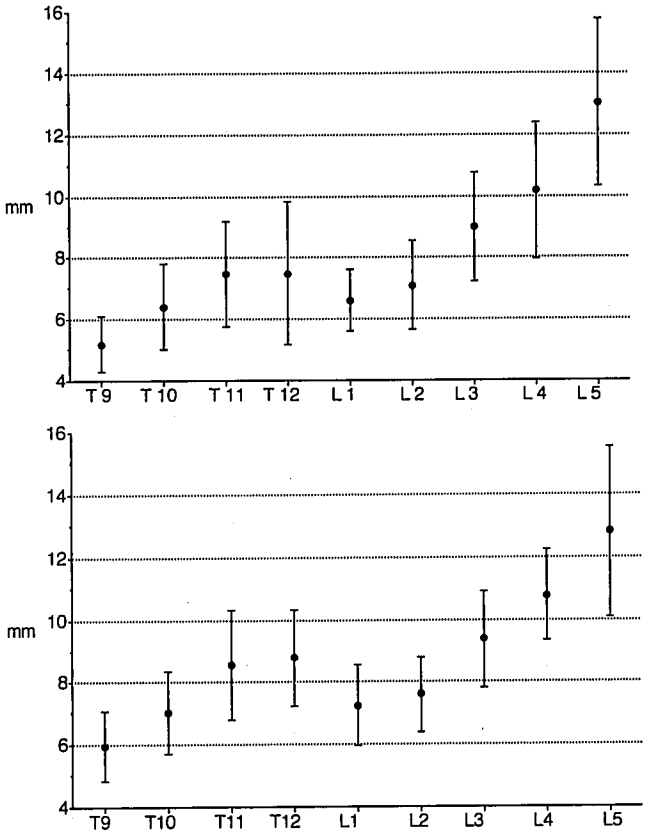


Figure 3. **A**, Female pedicle width (A) from T9 to L5. **B**, Male pedicle width (A) T9 to L5.

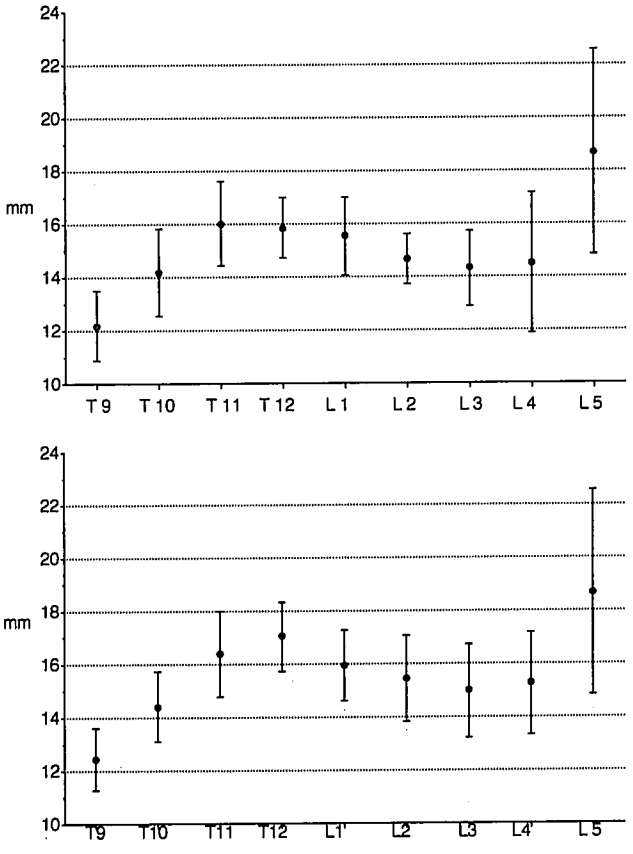


Figure 4. **A**, Female pedicle height (B) T9 to L5. **B**, Male pedicle height (B) T9 to L5.

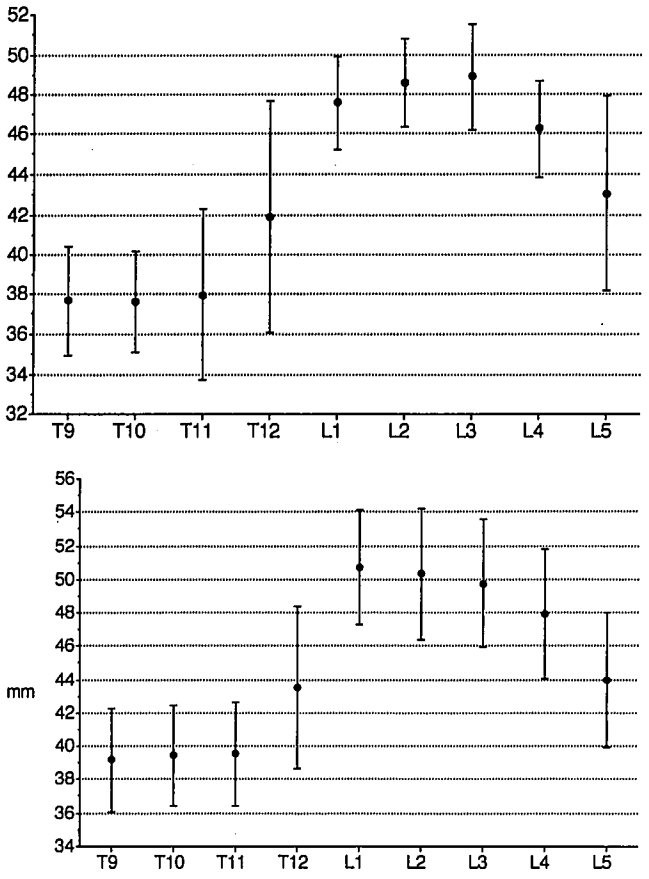


Figure 5. **A,**. Female pedicle depth (LI) T9 to L5. **B,** Male pedicle depth (LI) T9 to L5.

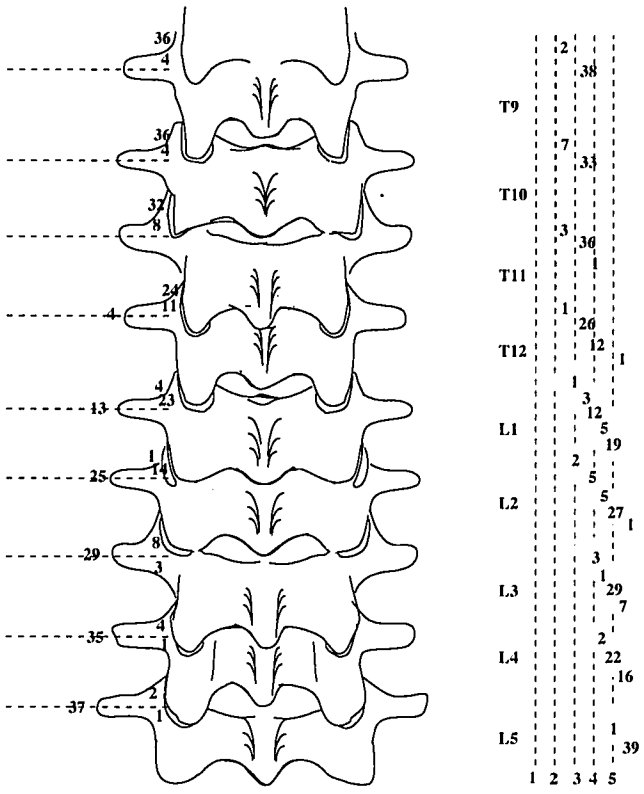


Figure 6. Relation of pedicle center to transverse process (on left). Position becomes more consistently at the mid-transverse process as one proceeds caudally. Relation of pedicle center to facet zones. Caudal progression demonstrates increasing deviation from the lateral facet joint. All but one center at L5 are lateral to the facet joint by one facet segment.

Comparison to Previous Data

Using data published by Zindrick et al,[10,11] Berry et al,[1] and Olsewski et al,[5] comparison of means of comparable measurements was performed. There were significant differences between pedicle widths reported by Zindrick and our recorded pedicle widths at all levels except T11. Pedicle length parallel to midline was also significantly different comparing Zindrick's data[10,11] and our own. When compared with data presented by Berry et al,[1] however, there were few differences of significance in relation to pedicle size except at the L5 level where pedicle width was significantly smaller in Berry's sample[1] compared with our own. Olsewski[5] reported significantly larger pedicle widths in males at all lumbar levels and in females at L1, L4, and L5. Pedicle height was somewhat more comparable in the females with significant differences at L2 and L5. Male pedicle height was similar at L2 (Table 2). General trends for pedicle measurement throughout all the different reports is of gradual increase in size to the L1 region, a sharp decline in size, and then a gradual increase in pedicle size, but decrease in vertebral depth from the mid-lumbar spine distally.

This population of specimens was measured in a similar manner to those done by Berry et al.[1] Figures that they report in their paper agree closely with our figures. Zindrick et al[10,11] used a radiographic technique that they thought did not contribute to magnification of their measurements. Their values are consistently and significantly different when compared with ours. Their pedicle widths are larger and their pedicle lengths are shorter. Although they describe similar points of assessment on the vertebrae, it may be that subtle differences in positions measured can account for these differences. Olsewski used a similar technique to ours to measure their values and arrived at significantly different sizes of

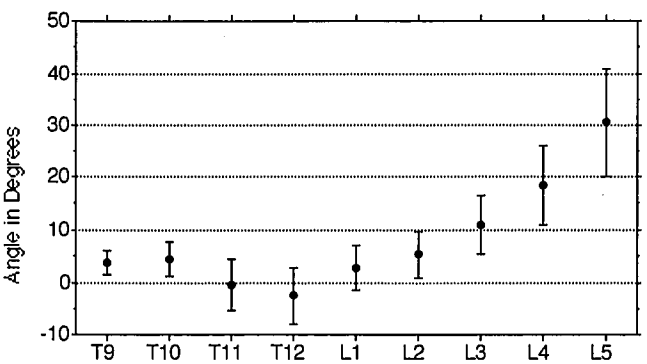


Figure 7. Inclination of pedicle from midline.

pedicles compared to our study. This was less marked in the female specimens but even so, three of five levels were significantly larger in their study. These differences may be a result of differences in population morphology with orientals having smaller vertebral sizes compared with the white population.

■ Conclusion

Pedicle width has some significant differences between males and females in the lower thoracic and upper lumbar spine. Care should be taken in these regions to ensure that pedicle screw diameter is appropriate for pedicle size. Position of pedicle starting points migrate laterally and approach the mid-transverse process line as one goes distally in the spine. Values of measurements may differ as a result of measurement methods and measurement populations.

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